Problem Statement:

Given an integer array nums, move all the even integers at the beginning of the array followed by all the odd integers.

Return *any array that satisfies this condition*.

Example 1:

Input: nums = [3,1,2,4]

Output: [2,4,3,1]

Explanation: The outputs [4,2,3,1], [2,4,1,3], and [4,2,1,3] would also be accepted.

Example 2:

Input: nums = [0]

Output: [0]

Constraints:

- **1 <= nums.length <= 5000**
- **0 <= nums[i] <= 5000**

Program:

```
import java.util.*;


public class Solution {
   public static int[] sortArrayByParity(int[] nums) {
      int[] result = new int[nums.length];
      int index = 0;


      // First add even numbers
      for (int num : nums) {
         if (num % 2 == 0) {
            result[index++] = num;
         }
      }
```

```
        // Then add odd numbers
        for (int num : nums) {
            if (num % 2 != 0) {
                result[index++] = num;
            }
        }

        return result;
    }

    public static void main(String[] args) {
        int[] nums = {3, 1, 2, 4};
        int[] output = sortArrayByParity(nums);

        // Print the result
        System.out.println(Arrays.toString(output));
    }
}
```